The pillars of quantitative portfolio management

SAA, Total Portfolio Approach, risk parity: how a portfolio is actually built.

SHARPE RATIO

$$\text{Sharpe} = (R_p - R_f) / \sigma_p$$ — excess return per unit of risk

STRATEGIC ASSET ALLOCATION

- Target split across asset classes over a long horizon (3-10 years).
- Based on expected returns/risks and cross-asset correlations.
- Reviewed periodically, but not adjusted to short-term market noise.

TOTAL PORTFOLIO APPROACH

- Manages the portfolio as a whole rather than in asset-class silos.
- Every investment is judged on its marginal contribution to the total portfolio.
- Popularized by large sovereign wealth funds and pension funds.

RISK MANAGEMENT

- Measuring and framing risk: VaR, tracking error, risk budget per sleeve.
- Distinction between absolute risk (capital loss) and relative risk (vs benchmark).
- Risk isn't eliminated — it's budgeted and monitored.

CONSTRUCTION & REBALANCING

- Turning a target allocation into an actual portfolio of instruments.
- Rebalancing: bringing the portfolio back to target weights after market drift.
- Trade-off between transaction costs and management discipline.

Who uses what?

PENSION FUNDS
SAA over a long horizon, calibrated to future retirement liabilities.

SOVEREIGN WEALTH FUNDS
Total Portfolio Approach to freely arbitrate across all opportunities.

HEDGE FUNDS
Risk parity and fine-grained risk budgets per strategy/factor.

INSURERS
Allocation constrained by liabilities and regulatory capital requirements.

KEY TAKEAWAYS

1. SAA sets the framework, rebalancing keeps it alive — without rebalancing discipline, the allocation drifts with the markets.
2. The Total Portfolio Approach breaks with siloed thinking: an investment is judged by its contribution to the whole portfolio, not in isolation.
3. Risk management isn't a final step but a continuous framework that constrains every allocation decision.

Comparison table

CRITERION	CLASSIC SAA	TOTAL PORTFOLIO APPROACH	RISK PARITY
Logic	By asset class	Unified whole portfolio	By risk contribution
Horizon	Long term (3-10 years)	Continuous, opportunistic	Continuous, frequently rebalanced
Complexity	Moderate	High (governance, data)	High (risk modeling)
Typical user	Pension fund, insurer	Sovereign fund, endowment	Hedge fund, multi-asset manager

Real-world examples

60/40 SAA

A classic 60% equities / 40% bonds portfolio, reviewed every 3 years based on long-term capital market assumptions.

NORWEGIAN SOVEREIGN FUND

Freely arbitrates between equities, bonds and real estate based on opportunities, without rigid silos — a Total Portfolio Approach example.

RISK PARITY PORTFOLIO

Equities, bonds and commodities weighted so each contributes equally to total risk, not to invested capital.

CALENDAR REBALANCING

A fund returns to its target weights every quarter, selling assets that outperformed and buying those that underperformed.

Quick glossary

EFFICIENT FRONTIER
Set of portfolios offering the best return for a given level of risk.

TRACKING ERROR
Standard deviation of the performance difference between a portfolio and its benchmark.

DRAWDOWN
Largest observed decline between a peak and a trough in portfolio value.

VAR
Maximum expected loss over a given horizon, at a set confidence level.

CORRELATION
Measure of how two assets move together — key to diversification.

RISK BUDGET
Share of total portfolio risk allocated to each strategy or asset class.